

Topic 1.2 - Rates of Change

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Calculate and interpret average rates of change from formulas, tables, graphs, and contexts.
- Compare rates on different intervals and estimate a rate at a point using nearby average rates.
- Simplify difference quotients and explain what the algebra reveals about changing rates.

Section A: Foundation and representation

Question 1

Algebra

No calculator

For $f(x) = 2x^2 - 7x + 4$, calculate the average rate on $[-1, 3]$.

Question 2

Table reasoning

No calculator

For points $(0, 6)$, $(2, 15)$, $(5, 21)$, $(9, 5)$, calculate each consecutive displayed average rate.

Section A continued

Question 3

Context

No calculator

Water use rises from 1.8 million L at 6 a.m. to 3.15 million L at noon. Find and interpret the average rate.

Question 4

Graph reasoning

No calculator

A graph contains $A(-3, 8)$, $B(1, -4)$, $C(6, 11)$. Find secant slopes AB and BC . Does the first slope prove the graph decreases everywhere on $[-3, 1]$?

Question 5

Local estimate

Calculator permitted

Given $p(5.98) = 72.421$ and $p(6.02) = 72.589$, estimate the rate at $x = 6$.

Section B: Application and reasoning**Question 6**

Symbolic reasoning

No calculator

For $h(x) = 4x^2 - x + 3$, simplify $\frac{h(x+k)-h(x)}{k}$, $k \neq 0$.

Question 7

Parameter

No calculator

For $F(x) = ax^2 + 5x - 2$, the average rate on $[1, 4]$ is 20. Find a .

Question 8

Comparison

No calculator

Two quantities both increase by 24. One change occurs over input length 3, the other over length 8. Compare rates.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student computes the rate from $(2, 7)$ to $(6, 19)$ as $(19 - 7)/6 = 2$. Diagnose and correct.

Question 10

Challenge

No calculator

Construct a quadratic with $q(0) = 4$, rate 1 on $[0, 2]$, and rate 7 on $[2, 4]$.

Question 11

Piecewise modeling

No calculator

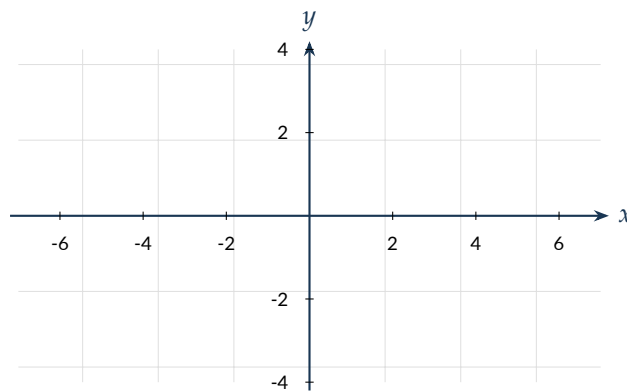
A piecewise-linear graph passes through $(0, 10)$, $(4, 22)$, $(7, 22)$, $(9, 8)$. Write each segment rule and identify greatest rate magnitude.

Question 12

Proof challenge

No calculator

Two continuous functions have the same values at 0, 2, 4. Must they be identical on $[0, 4]$? Relate your answer to average rates with those endpoints.



Section D: AP-style multiple choice

MCQ 1

No calculator

If $f(2) = 11$ and $f(8) = -7$, what is the average rate on $[2, 8]$?

- A. -18
- B. -3
- C. 3
- D. 18

MCQ 2

No calculator

A quantity changes at -4.5 kg/day for 12 days. What is net change?

- A. -54 kg
- B. -16.5 kg
- C. -0.375 kg
- D. 54 kg

MCQ 3

No calculator

What is the average rate of $p(x) = x^2 + 1$ on $[a, a + 3]$?

- A. $2a + 3$
- B. $2a + 6$
- C. $a^2 + 6a + 10$
- D. $3a + 3$

MCQ 4

No calculator

A rate at $x = 5$ is estimated using $[4.99, 5.01]$. Which statement is supported?

- A. It is automatically exact.
- B. It is a nearby secant rate.
- C. It equals the output at 5.
- D. The function must be linear nearby.

AP-style free response 1

Calculator permitted - 6 points

Mass $M(t)$, in grams, is measured at selected times.

t	0	2	5	7	9
$M(t)$	30	52	78	73	45

Also $M(4.8) = 77.6$, $M(5.2) = 78.0$.

A. Calculate and interpret the rate on $[0, 5]$.

B. Calculate the rate on $[5, 9]$ and compare magnitudes.

C. Estimate the rate at $t = 5$.

D. Explain why the data do not prove an absolute maximum at $t = 5$.

AP-style free response 2

No calculator - 6 points

Let $p(x) = ax^2 + bx + c$, $a \neq 0$.

A. Simplify $\frac{p(r+h)-p(r)}{h}$.

B. Show the average rate on $[u, v]$ is $a(u + v) + b$.

C. Rates on $[1, 3]$ and $[3, 5]$ are 4 and 12. Find a, b .

D. If $p(0) = 7$, write p .
